

Computational Thinking

## Discrete Mathematics

Number Theory

Topic 04 : Relations and Functions

Logic

### Lecture 04 : Function Operations

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Graphs and  
Networks

Autumn Semester, 2026/27

Collections

#### Outline

- Function Operations
- Inverse of a Function — existence conditions and derivation

Enumeration

Relations & Functions

# Outline

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|                                   |    |
|-----------------------------------|----|
| 1. Operations                     | 2  |
| 1.1. Function Equality            | 5  |
| 1.2. Add/Subtract/Multiply/Divide | 6  |
| 1.3. Function Composition         | 8  |
| 2. Function Inverse               | 11 |

# Functions — Where are we ?

At this point we have:

- defined what a function is (any process that generates exactly one output for each input)
- covered fundamental concepts (source, target, domain, image),
- covered properties (injective, surjective and bijective).

we want to discuss

- function operations — constructing new functions by adding/multiplying functions\* or by applying one function after another function.
- function inverse — finding function pairs that have the property that applying one after the other results in the original input.
- yet another graphical representation of functions — using 2D Cartesian graphs to represent functions.
- a library of useful functions in computing.

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\*These are a bigger deal in calculus than in discrete mathematics

# Evaluating Functions

Before we start combining functions, I want to make sure that you are happy with evaluating a function.<sup>†</sup>

## Example 1

Given the function  $f : x \mapsto 2x^2 - x + 3$ , evaluate

❶  $f(-a)$

❷  $f(2a)$

❸  $f(a + h)$

❹  $f(x + 5)$

❶  $f(-a)$

$$f(-a) = 2[-a]^2 - [-a] + 3 = 2a^2 + a + 3$$

❷  $f(2a)$

$$f(2a) = 2[2a]^2 - [2a] + 3 = 8a^2 - 2a + 3$$

❸  $f(a + h)$

$$f(a + h) = 2[a + h]^2 - [a + h] + 3 = 2a^2 + 4ah + 2h^2 - a - h + 3$$

❹  $f(x + 5)$

$$f(x + 5) = 2[x + 5]^2 - [x + 5] + 3 = 2x^2 + 10x - x + 48$$

<sup>†</sup>Simply use an extra set of brackets to ensure correct order of operations.

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# Function Equality

Two functions are equal if they have the same domain and the same rule/mapping.

## Definition 2 (Function Equality)

Let  $f$  and  $g$  be two functions. Then

$$f = g \quad \Longleftrightarrow \quad \underbrace{\text{Dom}(f) = \text{Dom}(g)}_{\text{same domain}} \quad \wedge \quad \underbrace{f(x) = g(x) \quad \forall x \in \text{Dom}(f)}_{\text{same rule}}$$

- Two functions that have different domains cannot be equal. For example,

$$f : \mathbb{Z} \rightarrow \mathbb{Z} : x \mapsto x^2 \quad \text{and} \quad g : \mathbb{R} \rightarrow \mathbb{R} : x \mapsto x^2$$

are **not** equal even though the rule that defines them is the same.

- However, it is not uncommon for two functions to be equal even though they are defined differently. For example

$$h : \{-1, 0, 1, 2\} \rightarrow \{0, 1, 2\} : x \mapsto |x|$$

and

$$k : \{-1, 0, 1, 2\} \rightarrow \{0, 1, 2\} : x \mapsto -\frac{x^3}{3} + x^2 + \frac{x}{3}$$

appear to be very different functions. However, they are equal because, domains are equal and  $h(x) = k(x)$  for all  $x \in \{-1, 0, 1, 2\}$ .

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# Function Addition/Subtraction/Multiplication/Division

I'm throwing these four operations together in the hope that you see that this is just notational convenience<sup>‡</sup>. You will cover these more formally in your *Calculus* module.

## Definition 3

Given two functions  $f : x \mapsto f(x)$  and  $g : x \mapsto g(x)$  then (informally) the

- sum function is

$$(f + g) : x \mapsto f(x) + g(x)$$

- difference function is

$$(f - g) : x \mapsto f(x) - g(x)$$

- product function is

$$(fg) : x \mapsto f(x)g(x)$$

- quotient function is

$$(f/g) : x \mapsto f(x)/g(x) \quad g(x) \neq 0$$

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<sup>‡</sup>What programmers call “syntax sugar”.

# Example 4

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Let  $f : x \mapsto x^4 - 16$  and  $g : x \mapsto |x| - 4$  Determine

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- ❷  $(fg)(2)$
- ❸  $\left(\frac{f}{g}\right)(2)$
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# Function Composition

## Definition 5 (Function Composition)

Let  $f : A \rightarrow B$  and  $g : B \rightarrow C$ . Then the composition of  $f$  followed by  $g$ , written  $g \circ f$  is a function from  $A$  into  $C$  defined by

$$(g \circ f)(x) = g(f(x))$$

which is read as “ $g$  of  $f$  of  $x$ ” or “ $g$  after  $f$  of  $x$ ”

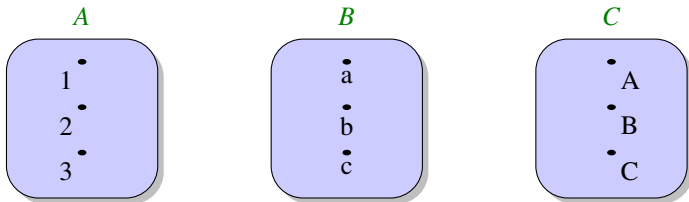
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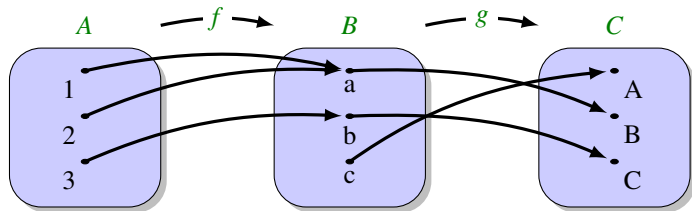
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$$f = \{(1, a), (2, a), (3, b)\}$$

$$g = \{(a, B), (b, C), (c, A)\}$$

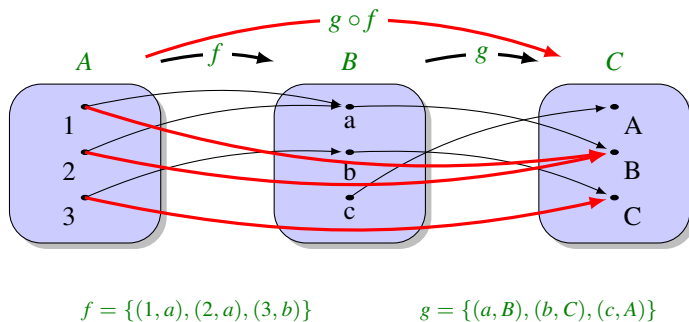
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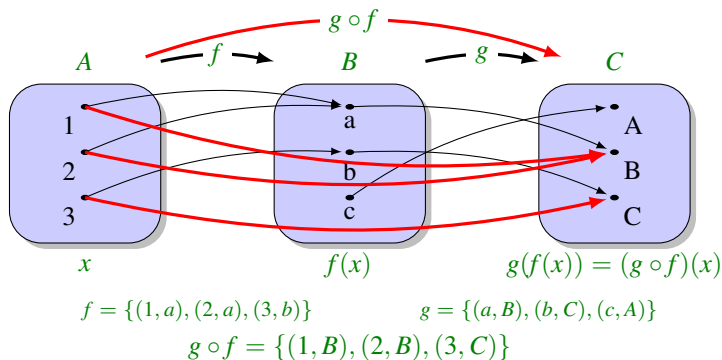
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# Example 6

## Example 6 (Function composition using formulae)

Consider functions  $f : \mathbb{R} \rightarrow \mathbb{R} : x \mapsto x^3$  and  $g : \mathbb{R} \rightarrow \mathbb{R} : x \mapsto 3x + 1$ . Then, construct functions  $g \circ f$  and  $f \circ g$ .

$g \circ f$

$$g \circ f : \mathbb{R} \rightarrow \mathbb{R} : x \mapsto g(f(x))$$

and since  $g(f(x)) = g(x^3) = 3[x^3] + 1$  we have

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$$f \circ g : \mathbb{R} \rightarrow \mathbb{R} : x \mapsto 27x^3 + 27x^2 + 9x + 1$$

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# Properties of Function Composition

While the previous example shows that we cannot change the order of functions in a function composition we are free to change the grouping ...

## Theorem 7 (Function composition is associative)

Given three function,  $f : A \rightarrow B$ ,  $g : B \rightarrow C$ , and  $h : C \rightarrow D$ , then

$$h \circ (g \circ f) = (h \circ g) \circ f$$

- This result means that no matter how the functions in the expression  $h \circ g \circ f$  are grouped, the final image of any element of  $x \in A$  is  $h(g(f(x)))$

Using function composition we can define repeated application of functions<sup>§</sup> ...

## Definition 8 (“Powers” of Functions)

Let  $f : A \rightarrow A$ .

- $f^1 = f$ ; that is,  $f^1(a) = f(a)$ , for  $a \in A$ .
- For  $n \geq 1$ ,  $f^{n+1} = f \circ f^n$ ; that is,  $f^{n+1}(a) = f(f^n(a))$  for  $a \in A$ .

<sup>§</sup>Take care of notation here:  $f^2(x) \neq (f(x))^2$ , etc.

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# Inverse of a Function

## Definition 9 (Inverse of a Function)

Let  $f : A \rightarrow B$ . If there exists a function  $g : B \rightarrow A$  such that

$$(g \circ f)(x) = x \quad \forall x \in A \quad \text{and} \quad (f \circ g)(x) = x \quad \forall x \in B$$

then  $g$  is called the inverse of  $f$  and is denoted by  $f^{-1}$ , read “ $f$  inverse”.

- Notice that in the definition we refer to “the inverse” as opposed to “an inverse” because, if the inverse exists it is unique.
- The inverse effectively “undoes” the effect of  $f$ .

$$\text{If } f(a) = b \text{ then } f^{-1}(b) = a$$

- The inverse of  $f$  exists if and only if  $f$  is bijective, i.e.,  $f$  is one-to-one and onto.
- Existence of a function inverse is fundamental to cryptography, lossless compression, relational databases, communication protocols, etc.
- Existence implies nothing about the relative ease of obtaining  $f^{-1}$ , or if found the effort to compute  $f^{-1}(x)$ .

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- The inverse of  $f$  exists if and only if  $f$  is bijective, i.e.,  $f$  is one-to-one and onto.
- Existence of a function inverse is fundamental to cryptography, lossless compression, relational databases, communication protocols, etc.
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# Inverse of a Function

## Definition 9 (Inverse of a Function)

Let  $f : A \rightarrow B$ . If there exists a function  $g : B \rightarrow A$  such that

$$(g \circ f)(x) = x \quad \forall x \in A \quad \text{and} \quad (f \circ g)(x) = x \quad \forall x \in B$$

then  $g$  is called the inverse of  $f$  and is denoted by  $f^{-1}$ , read “ $f$  inverse”.

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## Example 10

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On the set  $A = \{0, 1, 2, 3, 4\}$  the functions

$$f : A \rightarrow A : x \mapsto -\frac{5}{6}x^4 + \frac{20}{3}x^3 - \frac{50}{3}x^2 + \frac{83}{6}x$$

and

$$g : A \rightarrow A : x \mapsto 2x \bmod 5$$

are inverse functions.

# Example 10

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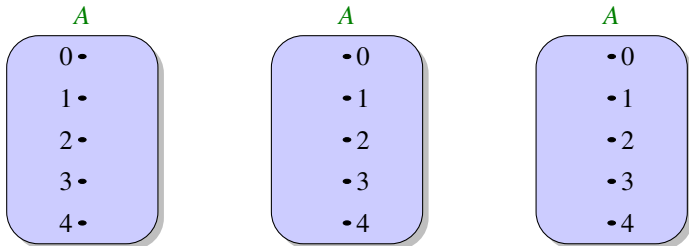
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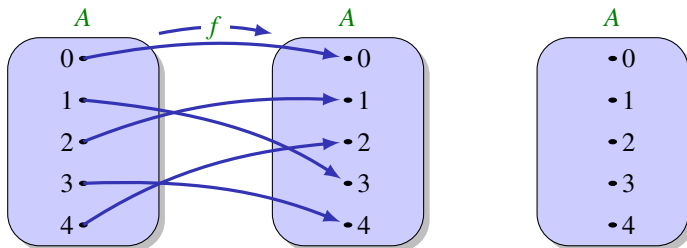
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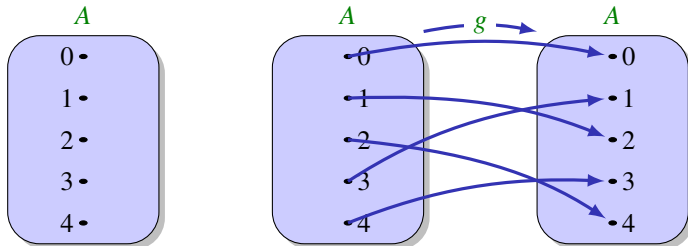
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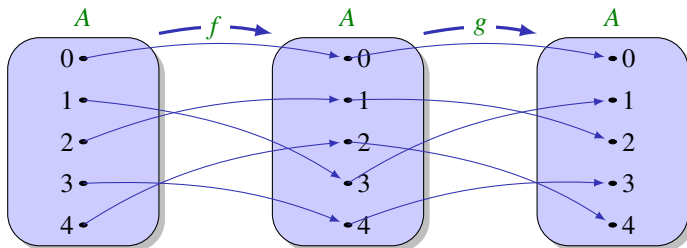
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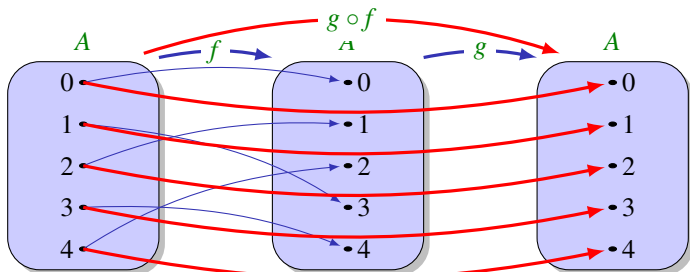
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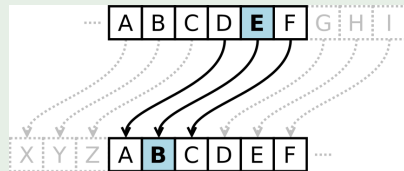


# Example — Caesar Cipher

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## Example 11 (Caesar Cipher)

The Caesar cipher, also known as a **shift cipher**, is one of the simplest forms of encryption. It is a substitution cipher where each letter in the original message (called the plaintext) is replaced with corresponding letter at a fixed shift<sup>¶</sup> in the alphabet with wrap around.



Decrypting with shift of 3.

If  $n$  is the required shift, and we have functions to map letters to/from integers such that 'A'  $\leftrightarrow$  0, 'B'  $\leftrightarrow$  1, ..., 'Z'  $\leftrightarrow$  25 then we have inverse function pair

$$E_n(x) = (x + n) \bmod 26$$

and

$$D_n(x) = (x - n) \bmod 26$$

In other words,  $(D_n \circ E_n)(x) = x$

<sup>¶</sup>Apparently Caesar used to prefer an offset of 3 letters, and would shave slaves' head, tattoo encrypted message, wait till hair regrows and then send "message".

# Example — Caesar Cipher

## Application

Caesar's used<sup>‡</sup> a shift of 3 so had encrypt/decrypt inverse pair  $E_3$  and  $D_3$ ,



The following message was encrypted using  $E_3$

*V H Q G P R U H I R R G*

Decrypt the message

-----

<sup>‡</sup>Security-wise, this is worse than useless, and has not been used since the 16<sup>th</sup> century, but a shift of 13 was (is?) popular in usenet newsgroups when posting offensive content. Google “ROT13”

# Example — Caesar Cipher

## Implementation

If  $n$  is the required shift, then using the `ord` and `chr` functions in Python\*\* we have inverse function pair

$$E_n(c) = \text{chr} \left( \underbrace{\left( \underbrace{(\text{ord}(c) - \text{ord}('A'))}_{\text{get integer in range } 0 \dots 25} + n \right) \bmod 26}_{\text{apply shift}} + \text{ord}('A') \right)$$

$\underbrace{\hspace{10em}}_{\text{apply wrap around}}$   
 $\underbrace{\hspace{15em}}_{\text{Add back ASCII offset}}$   
 $\underbrace{\hspace{20em}}_{\text{convert back to uppercase character}}$

and decrypt function

$$D_n(c) = \text{chr} \left( \left( (\text{ord}(c) - \text{ord}('A') + (26 - n)) \bmod 26 \right) + \text{ord}('A') \right) = E_{26-n}(c)$$

\*\*These functions map to/from ASCII values, so we have 'A' ↔ 65, 'B' ↔ 66, ..., 'Z' ↔ 90

# Example — Caesar Cipher

## Functions to Encode/Decode Caesar Cipher

```
1 def shift(n, x):
    return (x+n) % 26

def encrypt(n,message):
    result = ""
    for c in message:
        if 'A'<=c<='Z':
            result += chr(shift(n, ord(c)-ord('A')) + ord('A'))
        else:
            result += c
    return result
```

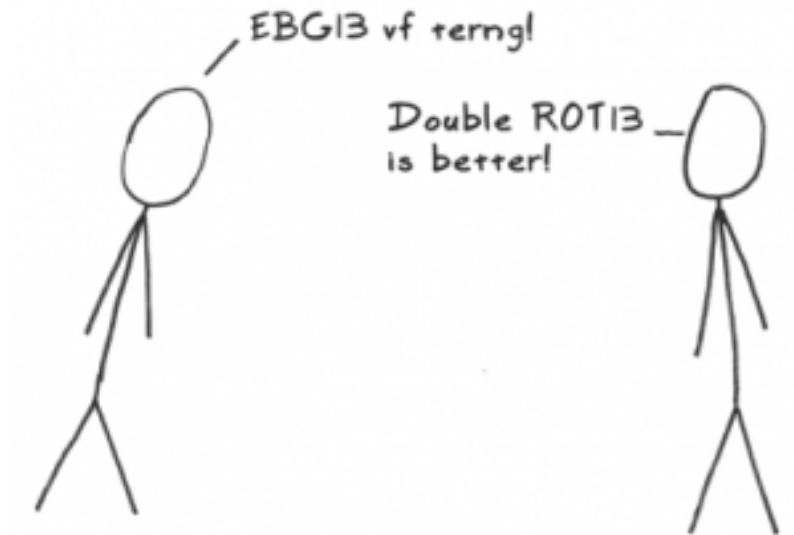
## Sample Usage

```
2 plaintext = "ATTACK AT DAWN"
  ciphertext = encrypt(3, plaintext)
  test = decrypt(3, ciphertext)

print ("Plaintext = ", plaintext)
print ("Ciphertext = ", ciphertext)
print ("test      = ", test)
```

```
Plaintext = ATTACK AT DAWN
Ciphertext = DWWDFN DW GDZQ
test      = ATTACK AT DAWN
```

## ROT13



# Review Exercises 1 (Function Inverse)

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## Question 1:

Let  $A = \{1, 2, 3\}$ . Define  $f : A \rightarrow A$  by  $f(1) = 2, f(2) = 1$ , and  $f(3) = 3$ . Find  $f^2, f^3, f^4$  and  $f^{-1}$ .

## Question 2:

Let  $f, g$ , and  $h$  all be functions from  $\mathbb{Z}$  into  $\mathbb{Z}$  defined by  $f(n) = n + 5, g(n) = n - 2$ , and  $h(n) = n^2$ . Define:

(a)  $f \circ g$

(b)  $f^3$

(c)  $f \circ h$

## Question 3:

Define  $s, u$ , and  $d$ , all functions on the set of integers,  $\mathbb{Z}$ , by  $s(n) = n^2, u(n) = n + 1$ , and  $d(n) = n - 1$ . Determine:

(a)  $u \circ s \circ d$

(b)  $s \circ u \circ d$

(c)  $d \circ s \circ u$

## Question 4:

Define the following functions on the integers by  $f(k) = k + 1, g(k) = 2k$ , and  $h(k) = \lceil k/2 \rceil$

(a) Which of these functions are one-to-one?

(b) Which of these functions are onto?

(c) Express in simplest terms the compositions  $f \circ g, g \circ f, g \circ h, h \circ g$ , and  $h^2$ ,